

AKASH S

AI ENGINEER | APPLIED AI | GENERATIVE AI | MACHINE LEARNING

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI Engineer shipping production systems for automotive supply chain intelligence, document AI, RAG, speech recognition, and computer vision, from a multi-tenant supply chain exception platform to fine-tuned speech models. Owns end-to-end delivery across data ingestion, retrieval, agentic workflows, evaluation, FastAPI services, React applications, and AWS deployment. Strong in Python, SQL, LLM orchestration, multi-tenant architecture, testing, and observability.

TECHNICAL SKILLS

Generative AI: LLMs, RAG, AI agents, agentic workflows, prompt engineering, embeddings, vector search, LangChain, LangGraph, Amazon Bedrock, OpenAI, Gemini, MCP

Machine Learning: Model evaluation, fine-tuning, LoRA, Transformers, Whisper, scikit-learn, computer vision, OCR, anomaly and risk detection

Programming and Applications: Python, SQL, TypeScript, JavaScript, FastAPI, Pydantic, REST APIs, Celery, React, Next.js, Redux, Node.js

Data, Cloud and MLOps: Pandas, NumPy, PostgreSQL, SQLAlchemy, ChromaDB, MemoryDB, S3, SageMaker, ECS, ECR, EventBridge, Azure, Docker, CI/CD

Quality and Security: Pytest, GitHub Actions, LangSmith, Weights & Biases, Sentry, JWT, Keycloak, RBAC, feature flags, Git

AI Developer Tools: Claude Code, Codex, Cursor, GitHub Copilot

PROFESSIONAL EXPERIENCE

Rokkun

Bengaluru, India

AI Engineer

Jun 2026 - Present

- Built and shipped a supply chain platform that catches EDI, demand, inventory, shortage, and expedite risks early and gives planners a clear, ready-to-act alert. Built with Python, FastAPI, Celery, PostgreSQL, and React.
- Made the system safe for many customers at once: each tenant sees only its own data through role-based access, and database migrations and alert delivery are safe to retry without creating duplicates.
- Designed an LLM pipeline that reads external OEM and industry signals, scores how relevant each one is, links it to the right parts, and shows the supporting evidence instead of overstating results when data is missing.
- Built the planner-facing decision workflow end to end, from scheduled data ingestion to the React and Redux screens, explaining why each part or vehicle program is flagged and capturing planner feedback.
- Shipped tariff and trade-document intelligence that finds official sources, parses PDFs with OCR, validates HTS codes and exemptions, and stores an audit-ready record of every decision; also simplified a fragile multi-agent RAG step into a leaner, more reliable one.

BGS Infotech

Bengaluru, India

AI Software Intern

Sep 2025 - Apr 2026

- Developed a liver tumor classification model using image processing and machine learning; results contributed to a co-authored medical AI research paper submitted for publication.
- Delivered client e-commerce features across requirements analysis, software implementation, testing, code review, and production deployment.
- Led adoption of AI-assisted developer tools to improve engineering productivity and code review practices across the team.

Prime Minds Consultancy

Bengaluru, India

AI Intern

Feb 2025 - May 2025

- Engineered a generative AI natural language query interface for Odoo ERP using RAG and LLM orchestration, enabling non-technical users to retrieve enterprise data in plain English.
- Designed grounded RAG pipelines integrating PostgreSQL and vector search, and automated Python-based data extraction and validation workflows.

AI PROJECTS

Nexa RAG AI | Enterprise Document Intelligence Platform [Project](#)

- Architected end-to-end ingestion and RAG for PDF, DOCX, and TXT files using MiniLM embeddings, ChromaDB, top-8 retrieval, confidence scoring, and citation-backed responses.
- Built LangGraph workflows for multi-document summarization, comparison, action extraction, and email drafting; delivered a FastAPI backend with JWT, SQLAlchemy, PostgreSQL, 15+ REST endpoints, and pluggable LLM providers.
- Containerized the API, PostgreSQL, ChromaDB, and React services with Docker Compose, persistent volumes, and environment-based configuration.

Hindi ASR Fine-Tuning | Whisper Small with LoRA [Project](#)

- Fine-tuned Whisper Small on Google FLEURS Hindi using LoRA, training approximately 0.8% of model parameters on a Kaggle T4 GPU.
- Reduced WER from 61.0% to 31.2% and CER from 30.7% to 11.2% on a 418-sample held-out test set; benchmarked Whisper Tiny, Base, Small, and Medium.
- Designed a FastAPI transcription deployment roadmap with JWT authentication, rate limiting, and ONNX optimization for efficient inference.

ForKrishi AI | Voice-First Agricultural Intelligence [Project](#)

- Built multilingual crop disease diagnosis using Gemini Vision, bounding-box overlays, and speech input and output in Kannada, Hindi, and English.
- Implemented an offline-capable PWA with service-worker caching, geolocation-based KVK discovery, and voice-enabled Mandi price search for field use.
- Developed advisory image export for WhatsApp sharing and deployed the application on Vercel using Next.js serverless routes.

Druva AI | Streaming Developer Chat Assistant [Project](#)

- Built a real-time developer assistant with Groq-powered Server-Sent Events streaming, asynchronous generators, and progressive React state updates for responsive conversational workflows.
- Implemented speech-to-text input, persistent browser chat history, mathematical expression rendering, context-aware prompting, and responsive light, dark, and OLED themes.

EDUCATION

Visvesvaraya Technological University

Bengaluru, India

Bachelor of Engineering, Information Science and Technology | CGPA: 8.38/10

Aug 2021 - Jun 2025

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Machine Learning, Computer Networks

ACHIEVEMENTS

- Winner, Zero Boundaries Hackathon: Built an AI multiplayer coding platform with automated code review and real-time debugging suggestions.
- Co-author, Liver Tumor Classification Research Paper: Applied machine learning and image processing to medical imaging data; paper submitted for publication.
- Cricket Champion, SKIT Premier League: Led the team through a competitive tournament, demonstrating ownership, collaboration, and calm decision-making under pressure.

CERTIFICATIONS

- Generative AI Engineering and Fine-Tuning Transformers, IBM, 2026
- Fundamentals of AI Agents Using RAG and LangChain, IBM, 2026
- Microsoft Azure Cloud Services, Microsoft, 2025